KAIZEN MATHS

Lesson starter

Basic Differentiation of Polynomials

Basic Power Rule

Name _____

S1

Differentiate with respect to x.

$f(x) = x^3$

S2

Differentiate with respect to x.

$f(x) = 2x^3$

S3

Differentiate with respect to x.

$f(x) = 2x$

S4

Differentiate with respect to x.

$f(x) = x^7$

Show your working

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ALG-E739-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Basic Differentiation of Polynomials

Basic Power Rule

Name _____

E1

Differentiate with respect to x.

$f(x) = 7x^5$

E2

Differentiate with respect to x.

$f(x) = x$

E3

Differentiate with respect to x.

$f(x) = x^4$

Show your working

KAIZEN MATHS

Exit ticket

Basic Differentiation of Polynomials

Basic Power Rule

Name _____

E1

Differentiate with respect to x.

$f(x) = 7x^5$

E2

Differentiate with respect to x.

$f(x) = x$

E3

Differentiate with respect to x.

$f(x) = x^4$

ALG-E739-L1 · E

Show your working

KAIZEN MATHS

Exit ticket

Basic Differentiation of Polynomials

Basic Power Rule

Name _____

E1

Differentiate with respect to x.

$f(x) = 7x^5$

E2

Differentiate with respect to x.

$f(x) = x$

E3

Differentiate with respect to x.

$f(x) = x^4$

ALG-E739-L1 · E

Show your working

KAIZEN MATHS

Exit ticket

Basic Differentiation of Polynomials

Basic Power Rule

Name _____

E1

Differentiate with respect to x.

$f(x) = 7x^5$

E2

Differentiate with respect to x.

$f(x) = x$

E3

Differentiate with respect to x.

$f(x) = x^4$

ALG-E739-L1 · E

Show your working

Teacher answers & working

Basic Differentiation of Polynomials · Basic Power Rule

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S1

Differentiate with respect to x.

$f(x) = x^3$

Answer:

$f'(x) = 3x^2$

Working:

Use the power rule.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Combine the term derivatives.

S2

Differentiate with respect to x.

$f(x) = 2x^3$

Answer:

$f'(x) = 6x^2$

Working:

Use the power rule.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Combine the term derivatives.

S3

Differentiate with respect to x.

$f(x) = 2x$

Answer:

$f'(x) = 2$

Working:

Use the power rule.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Combine the term derivatives.

Teacher answers & working

Basic Differentiation of Polynomials · Basic Power Rule

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S4

Differentiate with respect to x.

$f(x) = x^7$

Answer:

$f'(x) = 7x^6$

Working:

Use the power rule.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Combine the term derivatives.

E1

Differentiate with respect to x.

$f(x) = 7x^5$

Answer:

$f'(x) = 35x^4$

Working:

Use the power rule.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Combine the term derivatives.

E2

Differentiate with respect to x.

$f(x) = x$

Answer:

$f'(x) = 1$

Working:

Use the power rule.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Combine the term derivatives.

x is x^1, so the derivative is 1.

Teacher answers & working

Basic Differentiation of Polynomials · Basic Power Rule

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E3

Differentiate with respect to x.

$f(x) = x^4$

Answer:

$f'(x) = 4x^3$

Working:

Use the power rule.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Combine the term derivatives.